Coronary Stenosis Detection From Angiogram Images

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Abstract

Background. Coronary artery disease (CAD) remains one of the leading causes of mortality worldwide, with coronary artery stenosis serving as one of its most critical manifestations. Coronary angiography is widely regarded as the gold standard for stenosis diagnosis; however, manual interpretation of angiogram images is time-consuming, highly dependent on expert experience, and susceptible to inter-observer variability. These challenges highlight the need for automated and interpretable computer-aided diagnostic systems capable of assisting clinicians in accurate coronary stenosis detection.

Methods. This study proposes a deep learning–based framework for automated coronary stenosis detection using coronary angiogram images. The proposed approach integrates transfer learning with the EfficientNet-B4 architecture for robust feature extraction and classification. The framework incorporates comprehensive preprocessing and augmentation strategies, including geometric and intensity transformations, to improve model generalization. Bounding box–based annotation processing is utilized to enhance spatial localization of stenotic regions. To address dataset imbalance, weighted sampling and focal loss with label smoothing are implemented during training. Model performance is evaluated using accuracy, precision, recall, F1-score, Receiver Operating Characteristic (ROC), and Area Under the Curve (AUC). Additionally, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to provide visual interpretability of model predictions.

Results. Experimental evaluation demonstrates that the proposed framework achieves strong classification performance in distinguishing stenotic and non-stenotic coronary angiogram images. The model shows high discriminative capability across multiple evaluation metrics, particularly in recall and F1-score, indicating effective detection of clinically significant stenosis cases. Confusion matrix analysis reveals balanced classification performance with reduced false negative predictions, while ROC-AUC analysis confirms robust separability between classes. Furthermore, Grad-CAM visualizations indicate that the model consistently focuses on clinically relevant coronary vessel regions during prediction, supporting the reliability and interpretability of the learned feature representations.

Discussion. The results indicate that EfficientNet-B4 transfer learning combined with augmentation strategies and imbalance-aware optimization provides an effective solution for coronary stenosis detection from angiogram images. The integration of Grad-CAM enhances transparency by providing visual explanations that support clinician interpretation and trust in automated predictions. Compared with conventional black-box approaches, the proposed framework not only achieves competitive predictive performance but also improves explainability for potential clinical deployment. Nevertheless, limitations related to dataset diversity and binary classification scope remain. Future research should investigate larger multi-center datasets, stenosis severity assessment, and integration with real-time clinical decision support systems to further improve robustness and practical applicability.

Introduction

Coronary artery disease (CAD) remains one of the leading causes of morbidity and mortality worldwide, contributing substantially to the global burden of cardiovascular disorders. One of the most critical manifestations of CAD is coronary artery stenosis, a pathological narrowing of coronary arteries caused by atherosclerotic plaque accumulation that restricts myocardial blood flow and may lead to severe complications, including angina pectoris, myocardial infarction, heart failure, and sudden cardiac death (Thompson et al., 2022; van der Harst & Verweij, 2022). Because of these potentially life-threatening consequences, early and accurate detection of coronary stenosis is essential for timely intervention and effective patient management.

Coronary angiography is widely recognized as the gold standard imaging modality for evaluating coronary artery stenosis because it enables direct visualization of coronary vessel structures and blood flow obstruction (Huang et al., 2026; Cong et al., 2023). Despite its diagnostic importance, interpretation of angiographic images remains a complex and time-consuming task that requires substantial clinical expertise. The assessment process is frequently affected by inter-observer variability, inconsistent image quality, and the complexity of

coronary vessel anatomy (Danilov et al., 2021; Ostovaneh et al., 2022). Furthermore, the increasing volume of cardiovascular imaging examinations imposes considerable workloads on cardiologists, potentially reducing diagnostic efficiency and consistency. These challenges emphasize the need for automated computer-aided diagnostic systems capable of assisting clinicians in detecting coronary stenosis accurately and efficiently.

Recent advances in artificial intelligence (AI), particularly deep learning, have significantly transformed the field of medical image analysis. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in disease classification, lesion detection, image segmentation, and diagnostic prediction across various medical imaging domains (Wolterink et al., 2022; Hesamian et al., 2022). In cardiovascular imaging, deep learning approaches have shown promising results for the analysis of coronary angiography and related imaging modalities, enabling automated identification of vascular abnormalities and clinically relevant anatomical structures (Nakanishi et al., 2021; Mousavi et al., 2022). Unlike conventional machine learning methods that rely heavily on handcrafted feature extraction, CNN-based architectures automatically learn hierarchical representations directly from image data, resulting in more scalable and robust diagnostic systems (Tajbakhsh & Gurudu, 2022).

However, applying deep learning to coronary angiography analysis presents several challenges. One major limitation is the relatively limited availability of annotated medical imaging datasets. Medical image annotation is expensive, time-consuming, and dependent on expert clinical knowledge, which restricts dataset scale compared with natural image datasets. Consequently, training deep neural networks from scratch may lead to overfitting and poor generalization performance. Transfer learning has therefore emerged as an effective strategy for leveraging generalized visual representations learned from large-scale datasets and adapting them to domain-specific medical imaging tasks (Tajbakhsh & Gurudu, 2022; Zheng et al., 2023).

Among various CNN architectures, EfficientNet has gained considerable attention due to its compound scaling strategy, which balances network depth, width, and input resolution to improve learning efficiency while maintaining computational scalability. EfficientNet-based models have shown strong performance in medical image classification tasks, particularly in high-resolution imaging applications where both predictive accuracy and computational efficiency are required (Li et al., 2024; Zheng et al., 2023). By utilizing transfer learning with EfficientNet-B4, models can effectively adapt generalized visual features to coronary angiogram analysis while reducing training complexity and computational cost.

Another important challenge in medical image classification is class imbalance. In coronary angiography datasets, stenotic samples are often underrepresented compared with non-stenotic cases. Such imbalance may bias prediction toward majority classes and reduce sensitivity to clinically significant abnormalities. In coronary stenosis detection, false negative

predictions are particularly critical because undetected stenosis may delay treatment and increase the risk of severe cardiovascular events. Several studies have therefore emphasized the importance of imbalance-aware optimization strategies, including weighted sampling and specialized loss functions, to improve minority-class learning and enhance classification sensitivity (Ramos et al., 2024; Shalbaf & Behnam, 2022).

In addition to predictive performance, interpretability has become an increasingly important requirement in healthcare AI systems. Deep learning models are frequently criticized as "black-box" systems because their internal decision-making processes are often difficult to interpret, potentially reducing clinician trust and limiting clinical deployment (Albahri et al., 2023). Explainable Artificial Intelligence (XAI) methods have consequently been introduced to improve transparency and validate prediction behavior. One widely adopted approach is Gradient-weighted Class Activation Mapping (Grad-CAM), which generates visual heatmaps highlighting image regions that contribute most strongly to model predictions. Such visual explanations are valuable for verifying whether models focus on clinically relevant coronary vessel regions during diagnosis (Oksuz et al., 2023).

Several previous studies have investigated deep learning approaches for coronary artery disease and stenosis detection; however, many existing methods remain limited by insufficient imbalance handling, limited localization capability, or lack of explainability integration. Some studies emphasize classification accuracy without providing clinically interpretable visual evidence, while others employ computationally intensive architectures that may limit practical deployment in healthcare environments with constrained computational resources (Cong et al., 2023; Zreik et al., 2022; Moreno et al., 2022).

To address these limitations, this study proposes a deep learning–based framework for automated coronary stenosis detection using coronary angiogram images. The proposed framework integrates transfer learning with EfficientNet-B4, comprehensive preprocessing and augmentation strategies, imbalance-aware optimization using weighted sampling and focal loss with label smoothing, as well as explainable AI through Grad-CAM visualization. In addition, bounding box–based annotation processing is incorporated to improve spatial localization of stenotic regions during learning. By combining high predictive performance with interpretability and computational efficiency, this research aims to contribute a robust and clinically relevant computer-aided diagnostic system for assisting cardiologists in coronary angiography interpretation.

Methods

To systematically address the task of coronary stenosis detection from angiogram images, this study adopts a comprehensive deep learning pipeline consisting of six sequential phases. Each phase is designed to ensure data quality, robust feature learning, optimal model

performance, and interpretability. The overall workflow begins with raw image acquisition and proceeds through preprocessing, transfer learning–based modeling, optimization, evaluation, and explainability analysis.

Figure 1. Proposed Research Pipeline

Phase 1: Data Acquisition and Annotation Processing

The first phase of the proposed framework focuses on the acquisition, organization, and annotation processing of coronary angiogram image data. The dataset used in this study consists of coronary angiography images collected from clinical imaging examinations and prepared for supervised deep learning–based stenosis detection. Unlike conventional image-level classification approaches, the updated framework incorporates bounding box–based annotations to localize stenotic regions within coronary vessels, enabling the model to learn both pathological appearance and spatial information simultaneously. The use of localization-based learning in coronary angiography has been shown to improve lesion-focused analysis and enhance detection capability by directing model attention toward diagnostically relevant vascular regions (Cong et al., 2023; Ramos et al., 2024).

Each angiogram image is associated with annotation metadata containing the coordinates of regions suspected to exhibit coronary stenosis. These annotations are utilized to generate bounding boxes that identify clinically relevant vascular regions. Bounding box localization allows the learning framework to focus on stenotic vessel structures while minimizing the influence of irrelevant background artifacts commonly present in angiographic images, such as imaging devices, textual overlays, or non-coronary anatomical structures. Similar annotation-guided localization approaches have been reported to improve stenosis detection and lesion characterization in coronary imaging studies (Cong et al., 2023; Moraes et al., 2022).

During the annotation processing stage, the coordinate information extracted from annotation files is converted into a structured format compatible with the deep learning pipeline. The bounding box parameters are standardized to ensure consistency across all samples regardless of image resolution differences. Images without valid annotation information or containing corrupted metadata are excluded from the dataset to maintain annotation quality and training reliability. Careful annotation quality control is essential because annotation inconsistencies can substantially affect model generalization and localization accuracy in medical imaging systems (Meng et al., 2023).

Following annotation processing, the dataset is categorized into two primary classes:

- Stenosis, representing coronary arteries with clinically significant narrowing.

- Non-stenosis, representing normal or non-significant narrowing conditions.

To improve dataset robustness, images are inspected for duplication, incomplete annotations, and inconsistent labeling. This quality control process is particularly important because inaccuracies in annotation may negatively affect learning stability and predictive reliability (Danilov et al., 2021).

The prepared dataset is subsequently divided into training, validation, and testing subsets. The training set is used to optimize model parameters, the validation set is used to monitor performance during training and prevent overfitting, and the testing set is reserved exclusively for final evaluation. The splitting process preserves class distribution across all subsets to reduce sampling bias and improve evaluation fairness.

Since coronary angiography datasets commonly exhibit class imbalance due to the lower prevalence of stenotic cases compared with normal samples, the class distribution is analyzed prior to training. Previous studies have highlighted that imbalance in coronary imaging datasets may reduce model sensitivity toward clinically important stenotic lesions and bias predictions toward majority classes (Mousavi et al., 2022; Jeddi et al., 2023). This imbalance is addressed in subsequent phases through weighted sampling strategies and focal loss optimization to ensure that minority stenosis samples receive sufficient attention during model learning.

By integrating image-level classification labels with bounding box localization information, this phase establishes a reliable and clinically meaningful foundation for the subsequent preprocessing, augmentation, and EfficientNet-based feature learning stages within the proposed coronary stenosis detection framework.

Phase 2: Pre-processing

After dataset acquisition and annotation processing, the next phase focuses on image preprocessing and augmentation to improve data consistency, model robustness, and generalization capability. Coronary angiogram images often exhibit substantial variability in image resolution, illumination intensity, contrast distribution, vessel visibility, and background artifacts due to differences in acquisition devices and imaging protocols. Such variability may negatively affect feature extraction and classification performance if not properly addressed (Li et al., 2024; Wang et al., 2025).

The first preprocessing step involves resizing all angiogram images into a standardized spatial resolution compatible with the EfficientNet architecture used in this study. Since the proposed framework utilizes EfficientNet-B4 as the backbone model, all images are resized according to the network input requirement while preserving important anatomical structures

within coronary vessels. Image standardization is necessary to ensure consistent tensor representation during batch processing and neural network training (Hesamian et al., 2022).

Following image resizing, pixel intensity normalization is applied to stabilize the optimization process and improve convergence during training. The normalization process transforms pixel values into a controlled numerical range, reducing the effect of illumination variation between angiogram samples. Similar preprocessing techniques have demonstrated improved convergence and discriminative vascular feature learning in coronary imaging studies (Moreno et al., 2022).

To improve generalization capability, extensive data augmentation techniques are incorporated during training. Data augmentation artificially increases dataset diversity by generating transformed variations of existing samples while preserving clinically relevant vessel characteristics. Several augmentation strategies are applied, including random horizontal flipping, rotation, scaling, zooming, and translation. These transformations simulate real-world variability in angiographic imaging and reduce the risk of overfitting to specific image orientations or vessel positions. Previous coronary imaging research has shown that augmentation significantly improves robustness and classification performance when training data are limited (Li et al., 2024; Ahmad & Lee, 2022).

In addition to geometric augmentation, intensity-based augmentation techniques are applied to improve robustness against variations in contrast and illumination. Adjustments in brightness and image intensity help the model learn invariant features under different angiographic acquisition conditions. Such preprocessing strategies are particularly important in medical imaging tasks where dataset size is relatively limited compared with large-scale natural image datasets (Hesamian et al., 2022).

The preprocessing pipeline also integrates bounding box–aware transformations to ensure consistency between image augmentation and annotation coordinates. Every geometric transformation applied to the image is simultaneously propagated to the corresponding bounding box coordinates to preserve localization accuracy. Without such synchronization, annotation misalignment may occur and negatively impact localization learning (Cong et al., 2023).

To further improve training efficiency, processed images and annotations are converted into tensor representations compatible with the PyTorch framework. Batch loading and preprocessing operations are optimized through data loader mechanisms to accelerate training while maintaining computational stability.

Overall, this preprocessing and augmentation phase plays a critical role in enhancing dataset quality, increasing sample diversity, and improving the robustness of the proposed EfficientNet-B4–based coronary stenosis detection framework.

Phase 3: Transfer Learning Architecture using EfficientNet-B4

The third phase focuses on deep feature extraction and classification using transfer learning based on the EfficientNet-B4 architecture. Transfer learning is employed to overcome limitations associated with relatively small annotated medical imaging datasets while leveraging feature representations learned from large-scale image datasets. Previous studies have demonstrated that transfer learning improves convergence efficiency and predictive performance in cardiovascular image analysis (Tajbakhsh & Gurudu, 2022; Zheng et al., 2023).

This study adopts EfficientNet-B4 as the backbone architecture due to its strong balance between classification performance and computational efficiency. EfficientNet utilizes compound scaling to systematically balance network depth, width, and input resolution, enabling improved performance with fewer parameters compared with conventional CNN architectures. Similar architecture efficiency has contributed to successful medical imaging applications requiring high-resolution feature learning (Wolterink et al., 2022; Hesamian et al., 2022).

The EfficientNet-B4 model is initialized using weights pre-trained on ImageNet. Pre-training enables the network to capture generalized visual representations including edges, textures, vessel contours, and structural patterns that can subsequently be adapted to coronary angiogram analysis. Transfer learning has been widely recognized as an effective strategy for coronary artery disease classification and stenosis detection under limited data conditions (Shalbaf & Behnam, 2022; Zheng et al., 2023).

To adapt EfficientNet-B4 to coronary stenosis detection, the original classification layers are replaced with a custom classification head consisting of:

- Global feature extraction from EfficientNet backbone
- Fully connected dense layers
- Dropout regularization layers
- Final sigmoid activation for stenosis probability prediction

The sigmoid activation function is selected because the task is formulated as binary classification involving stenosis and non-stenosis categories.

In addition to image-level classification, the framework incorporates bounding box–guided learning to improve spatial sensitivity toward stenotic regions. By integrating

localization information during preprocessing and augmentation, the EfficientNet backbone learns diagnostically relevant vascular representations while minimizing attention to irrelevant image background. Similar region-focused learning strategies have improved coronary stenosis localization and lesion interpretation in recent studies (Cong et al., 2023; Ramos et al., 2024).

To further improve learning stability, selective fine-tuning is applied during training. Early convolutional layers responsible for generic feature extraction remain partially frozen, while deeper layers responsible for domain-specific learning are fine-tuned using coronary angiogram data. This strategy helps preserve useful pre-trained knowledge while enabling adaptation to medical imaging characteristics (Tajbakhsh & Gurudu, 2022).

The architecture is implemented using the PyTorch framework, supporting transfer learning, GPU acceleration, and flexible optimization pipelines. Overall, the integration of EfficientNet-B4 transfer learning provides a robust foundation for coronary stenosis classification by combining efficient feature extraction, parameter optimization, and domain adaptation suitable for real-world clinical applications.

Overall, the integration of EfficientNet-B4 transfer learning provides a robust foundation for coronary stenosis classification by combining efficient feature extraction, parameter optimization, and domain adaptation. The architecture enables the proposed framework to learn complex vascular patterns while maintaining computational scalability suitable for real-world clinical applications

Phase 4: Modeling with Hyperparameter Tuning

The fourth phase of the proposed framework focuses on model training, optimization, and class imbalance handling to improve the classification performance of coronary stenosis detection. Since coronary angiography datasets commonly contain unequal distributions between stenosis and non-stenosis samples, specialized optimization strategies are necessary to prevent the model from becoming biased toward majority classes (Mousavi et al., 2022; Jeddi et al., 2023).

The training process is performed using the PyTorch deep learning framework with GPU acceleration to improve computational efficiency during forward and backward propagation. The dataset is divided into training and validation subsets, where the training set is used for parameter optimization and the validation set is utilized to monitor generalization performance throughout the training process. Batch-wise learning is implemented to stabilize memory usage and accelerate gradient computation.

The proposed framework employs the AdamW optimizer as the primary optimization algorithm due to its adaptive learning rate mechanism and improved regularization capability.

AdamW combines momentum-based optimization with decoupled weight decay regularization, enabling more stable convergence and better generalization performance compared to conventional stochastic gradient descent approaches. The optimizer updates model parameters iteratively by minimizing the classification loss between predicted outputs and ground truth labels.

To further improve convergence stability, a learning rate scheduling strategy is incorporated during training. The scheduler dynamically adjusts the learning rate according to validation performance, allowing the optimization process to converge more effectively while reducing the risk of overshooting local minima. Lower learning rates are gradually applied during later training stages to refine the learned feature representations.

One of the most important aspects of this phase is handling class imbalance within the coronary angiogram dataset. Since stenosis samples are less frequent than non-stenosis samples, the model may become biased toward predicting majority classes if conventional loss functions are used. To address this issue, the proposed framework integrates two complementary imbalance-handling strategies:

- Weighted sampling
- Focal loss with label smoothing

Weighted sampling ensures that minority stenosis samples are more frequently represented during batch generation. This prevents the training process from being dominated by normal samples and improves sensitivity toward clinically significant stenotic cases (Mousavi et al., 2022; Shalbaf & Behnam, 2022).

In addition, the framework utilizes a customized Focal Loss function to prioritize difficult samples during optimization. Unlike standard binary cross-entropy loss, focal loss reduces the contribution of easily classified samples and increases attention toward hard-to-classify stenosis cases. This mechanism is particularly beneficial in medical diagnosis tasks where false negative predictions may lead to severe clinical consequences (Mousavi et al., 2022; Zheng et al., 2023).

To further improve robustness and reduce overconfidence in predictions, label smoothing is integrated into the focal loss formulation. Label smoothing prevents the model from assigning excessively confident probabilities to predictions, thereby improving calibration and reducing overfitting.

The training process also incorporates several regularization techniques to improve model generalization. Dropout layers within the classification head randomly deactivate neurons during training to reduce feature co-adaptation and prevent memorization of training samples. Furthermore, early stopping is applied based on validation performance to terminate training

when no significant improvement is observed across multiple epochs. This prevents excessive optimization that may lead to overfitting (Tajbakhsh & Gurudu, 2022).

Throughout the training process, multiple performance metrics are continuously monitored, including training loss, validation loss, accuracy, precision, recall, and F1-score. These metrics provide insight into both optimization stability and clinical classification capability. Particular emphasis is placed on recall and F1-score because accurate detection of stenosis cases is clinically more important than achieving high overall accuracy alone (Jeddi et al., 2023; Mousavi et al., 2022).

By combining transfer learning, adaptive optimization, focal loss, weighted sampling, and regularization techniques, this phase enables the proposed EfficientNet-B4 framework to learn robust and clinically meaningful feature representations for coronary stenosis detection.

Phase 5: Model Evaluation and Performance Analysis

The fifth phase of the proposed framework focuses on evaluating the inference stability and predictive reliability of the EfficientNet-B4–based coronary stenosis detection model. Since cardiovascular image classification systems must deliver both high predictive robustness and clinical consistency, a multifaceted evaluation pipeline engineered with rigorous metrics is utilized to provide a comprehensive assessment of the trained algorithm's capability (Cong et al., 2023; Ramos et al., 2024).

After completing the three-phase progressive training loop, the best-optimized model checkpoint (captured via the highest validation Macro-F1 score progression best_model.pth) is retrieved. This checkpoint is systematically evaluated using the completely unseen Test Dataset—a partitioned array specifically isolated through the patient-wise zero leakage technique. This strict isolation strategy ensures that the generated numerical performance unconditionally reflects the model’s true generalization capability rather than memorization of clinical markers from the training set.

Unlike conventional evaluation methodologies predicting static images, this framework deployed an advanced verification sequence named Test-Time Augmentation (TTA). Instead of processing a single fixed dimension for each angiograph, the TTA pipeline automatically transforms each unseen testing sample into disparate augmented iterations applying diverse horizontal flips, multi-angle geometric rotations, and normalized resizing functions. A deterministic forward pass generates probability matrices across all five views. The overall predictions are then mathematically ensembled and averaged to stabilize the softmax probabilities. As a consequence, establishing diagnostic consensus from multi-angled predictions greatly diminishes susceptibility toward transient image noises or localized blur spots initially present under standard evaluation (Moreno et al., 2022; Li et al., 2024).

Subsequent to computing the optimized probability thresholds into binary deterministic predictions ("Stenosis" and "No Stenosis"), multiple quantitative analytics metrics from the scikit-learn diagnostic module mapped the testing results:

- Test Accuracy
- Per-Class Precision, Recall (Detection Rate), & Specificity
- Macro F1-Score & Weighted F1-Score
- Balanced Accuracy
- Receiver Operating Characteristic (ROC) Curve & Area Under the Curve (AUC)

While base Accuracy measures the gross proportion of correct test samples, it functions inadequately within clinically skewed demographics. Therefore, emphasis is placed heavily on extracting relative, domain-agnostic metric dimensions. Recall (Sensitivity) maps the capability of the algorithm to intercept patients bearing actual arterial constraints, wherein suppressed recall equivalents correspond immediately to elevated, fatal false-negatives (FN) miss instances in cardiovascular screening. The Macro-F1 Score harmonizes the evaluation by mathematically penalizing unbalanced predictive behaviors regardless of class frequency, producing a balanced arithmetic mean between detection reliability (Precision) and disease intercept rates (Recall). A complementary Balanced-Accuracy algorithm additionally calculates an unbiased accuracy proportion calibrated symmetrically for respective minor and major cohorts (Mousavi et al., 2022; Ramos et al., 2024).

Expanding past standalone algorithmic numbers, an extensive quadrant analysis yields structural prediction mappings through both absolute and percentage-normalized Confusion Matrices. Examining the absolute fractionality dividing:

- True Positives (TP) and True Negatives (TN)
- False Positives (FP) and False Negatives (FN)

allows analysts to critically expose bias orientation. Simultaneous with these charts, an analytical framework is coded to execute Visual Error Analytics, filtering explicitly incorrect predictions—rendering juxtaposed images to review confidence probability deltas and anatomically identify what visual obstructions triggered failure during misclassification cases (Danilov et al., 2021; Cong et al., 2023).

Subsequently, discrimination boundaries are computed probabilistically across changing gradient thresholds plotting the Relative Operating Characteristic (ROC) curve. Mapping the non-linear relationship tracking true-positive rates against false-positive constraints produces the integrated Area Under the Curve (AUC) score. A maximized scalar closer to 1.0 indicates

consistent pathological pattern dissociation and separability confidence across the binary spectrum mappings regardless of threshold variations (Ramos et al., 2024; Zreik et al., 2022).

Finally, convergence analytics trace and render historical model training metrics (Training vs. Validation Loss, Accuracy Progressions, and Macro/Balanced Acc Evolutions) extracted systematically out of the phase arrays spanning N epochs. Visualizing the trajectory separation explicitly ensures mapping alignment confirming the validation integrity without evidence of overt gradient shattering or extreme domain divergence (overfitting) phenomena resulting from WeightedRandomSampling minority-class compensations (Tajbakhsh & Gurudu, 2022).

Overall, through aligning deterministic ensembles, harmonized scoring mathematics, comprehensive validation plotting, and structural bias examination protocols, this analytical framework ensures the finalized angiographic detection system is both statistically valid and practically reliable prior to diagnostic interpretability integration protocols via Grad-CAM deployment pipelines.

Phase 6: XAI Validation (Explainable AI)

The final phase of the proposed framework focuses on model interpretability and visual explainability using Gradient-weighted Class Activation Mapping (Grad-CAM). In medical image analysis, high predictive performance alone is insufficient for clinical adoption because clinicians must also understand the reasoning behind the model's predictions. Deep learning models are often criticized as "black-box" systems due to their complex internal representations and limited transparency. Therefore, explainability mechanisms are essential to improve trust, interpretability, and clinical validation (Albahri et al., 2023; Oksuz et al., 2023).

To address this limitation, the proposed framework integrates Grad-CAM to visualize the spatial regions within coronary angiogram images that contribute most significantly to stenosis predictions. Grad-CAM is a gradient-based visualization technique that generates heatmaps by analyzing the gradients flowing through the final convolutional layers of a convolutional neural network. These gradients are used to identify the regions that most strongly influence the model's classification decisions (Oksuz et al., 2023; Albahri et al., 2023).

In this study, Grad-CAM is applied to the EfficientNet-B4 backbone after the inference process. When an angiogram image is passed through the trained network, Grad-CAM computes the gradients of the target class score with respect to the feature maps of the final convolutional layer. The resulting weighted activation maps are then combined to produce localization heatmaps highlighting the most influential image regions.

The generated heatmaps are subsequently superimposed onto the original coronary angiogram images to provide intuitive visual explanations of model behavior. Areas receiving higher activation intensity indicate regions that contribute strongly to stenosis classification. Since the framework incorporates bounding box annotations during training, the Grad-CAM visualization can also be qualitatively compared with annotated stenotic regions to assess whether the model focuses on clinically relevant vascular structures (Albahri et al., 2023).

The integration of Grad-CAM serves several important objectives within this study. First, it enhances transparency by enabling clinicians and researchers to understand the visual patterns learned by the model. Second, it provides a qualitative validation mechanism to verify whether the model attends to anatomically meaningful coronary regions rather than irrelevant background artifacts. Third, explainability analysis helps identify potential prediction errors, localization failures, or dataset biases that may affect clinical reliability (Oksuz et al., 2023; van der Harst & Verweij, 2022).

In addition to qualitative interpretation, Grad-CAM visualizations are useful for assessing the effectiveness of the proposed preprocessing and augmentation strategies. Accurate activation localization indicates that the model successfully learns diagnostically meaningful vascular representations despite variations in imaging conditions, contrast distribution, and vessel orientation.

The explainability phase also contributes to improving trustworthiness in AI-assisted medical diagnosis systems. By providing visual evidence supporting classification decisions, Grad-CAM helps bridge the gap between automated deep learning predictions and clinician interpretation. This is particularly important in high-risk healthcare environments where explainability and accountability are essential for practical deployment (Albahri et al., 2023; van der Harst & Verweij, 2022).

Overall, the integration of Grad-CAM enhances the proposed coronary stenosis detection framework by combining strong predictive performance with interpretable decision-making. The explainability analysis demonstrates that the EfficientNet-B4–based model not only achieves accurate classification results but also focuses on clinically relevant coronary vessel regions, thereby supporting its potential application as a reliable computer-aided diagnostic tool.

Results and Discussion

Classification Performance Analysis

The proposed EfficientNet-B4–based framework was evaluated using the testing dataset consisting of unseen coronary angiogram images to assess its capability in detecting coronary stenosis. The evaluation results demonstrate that the model achieved strong performance in distinguishing stenotic and non-stenotic coronary artery conditions. The integration of transfer learning, augmentation strategies, focal loss optimization, and weighted sampling contributed significantly to improving classification robustness and sensitivity toward clinically significant stenosis cases.

The overall classification performance was assessed using multiple quantitative metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The results indicate that the proposed framework achieved stable and balanced performance across all evaluation metrics. In particular, the recall and F1-score values demonstrate the model's effectiveness in identifying stenotic cases while minimizing missed diagnoses. This outcome is particularly important in medical imaging applications because false negative predictions may delay treatment and increase cardiovascular risk.

The confusion matrix analysis further illustrates the classification behavior of the model. The majority of stenosis and non-stenosis samples were classified correctly, indicating that the proposed framework successfully learned discriminative vascular features from coronary angiogram images. Although several misclassifications were still observed, the number of false negatives remained relatively low, suggesting that the imbalance-handling strategies implemented during training effectively improved sensitivity toward minority stenosis samples.

The ROC curve analysis also demonstrates strong discriminative capability across varying decision thresholds. The high AUC value indicates that the model can reliably separate stenotic and non-stenotic images with consistent prediction performance. This result confirms the effectiveness of the EfficientNet-B4 architecture in extracting meaningful coronary vessel representations from angiographic data.

Effect of Data Augmentation and Imbalance Handling

The incorporation of extensive preprocessing and augmentation strategies significantly improved model generalization performance. Coronary angiogram images commonly exhibit variations in vessel orientation, illumination, contrast distribution, and acquisition conditions. Without augmentation, the model may overfit to specific image characteristics and perform poorly on unseen data.

The augmentation pipeline introduced geometric and intensity variations that simulated realistic clinical imaging conditions. As a result, the model became more robust to positional and visual variability within coronary angiograms. This improvement is reflected in the stable

validation performance observed during training and the reduced gap between training and validation metrics.

In addition, the implementation of weighted sampling and focal loss optimization successfully addressed the class imbalance problem commonly encountered in medical datasets. Conventional loss functions tend to bias predictions toward majority classes, reducing sensitivity toward stenosis cases. By emphasizing difficult and minority samples, focal loss improved the model's ability to detect clinically important stenotic regions. The improvement in recall and F1-score demonstrates the effectiveness of this imbalance-handling strategy.

The integration of label smoothing also contributed to reducing prediction overconfidence and improving calibration stability during optimization. This combination of focal loss and label smoothing enhanced generalization performance and reduced overfitting tendencies throughout the training process.

Transfer Learning Effectiveness

The use of transfer learning with EfficientNet-B4 provided substantial advantages in feature extraction and optimization efficiency. Since coronary angiography datasets are relatively limited in size compared to natural image datasets, training a deep convolutional neural network from scratch may lead to poor convergence and overfitting.

By utilizing ImageNet pre-trained weights, the proposed framework benefited from generalized low-level visual feature representations such as edges, textures, and structural patterns. Fine-tuning the deeper layers enabled the model to adapt these generalized representations to domain-specific coronary vessel characteristics. This transfer learning strategy accelerated convergence and improved overall classification performance.

EfficientNet-B4 also demonstrated strong computational efficiency compared with larger conventional CNN architectures. The compound scaling mechanism enabled the network to maintain high feature representation quality while controlling parameter growth. This balance between performance and computational efficiency is important for future deployment in real-time clinical environments.

Grad-CAM Explainability Analysis

To improve interpretability, Grad-CAM visualization was applied to analyze the spatial attention behavior of the proposed model. The resulting heatmaps indicate that the EfficientNet-B4 framework consistently focused on clinically relevant coronary vessel regions

during prediction. High activation areas corresponded closely with annotated stenotic regions identified within the angiogram images.

The Grad-CAM analysis demonstrates that the model learned meaningful pathological patterns rather than relying on irrelevant image artifacts or background noise. This finding supports the validity of the proposed preprocessing pipeline and bounding box–guided learning strategy. By incorporating localization information during training, the model became more spatially sensitive to stenotic vessel structures.

The explainability results are particularly important for clinical applications because they provide visual justification for automated predictions. Clinicians can verify whether the model attends to anatomically meaningful regions before accepting AI-assisted diagnostic recommendations. This transparency increases trustworthiness and supports the integration of deep learning systems into real-world healthcare workflows.

Limitations and Future Work

Despite the promising performance achieved by the proposed framework, several limitations remain. First, the dataset size is relatively limited compared to large-scale medical imaging repositories, which may affect model generalization across diverse patient populations and imaging devices. Second, the current framework focuses on binary stenosis classification and does not yet estimate stenosis severity levels or lesion complexity.

Future research should investigate the use of larger multi-center coronary angiography datasets to improve robustness and domain generalization. Additional studies may also explore:

- Multi-class stenosis severity classification,
- Real-time lesion localization,
- Hybrid CNN-transformer architectures,
- Self-supervised pretraining approaches, and
- Integration with clinical decision support systems.

Furthermore, quantitative explainability evaluation methods may be incorporated to complement Grad-CAM visualization analysis and provide deeper interpretability assessment.

Overall, the experimental results demonstrate that the proposed EfficientNet-B4–based framework provides an effective and interpretable solution for automated coronary stenosis detection from angiogram images. The combination of transfer learning, imbalance-aware optimization, augmentation strategies, and explainable AI contributes to a robust computer-aided diagnostic system with strong potential for clinical application.

Conclusions

This study proposed a deep learning–based framework for automated coronary stenosis detection using coronary angiogram images. The proposed system integrates transfer learning with the EfficientNet-B4 architecture, comprehensive preprocessing and augmentation strategies, imbalance-aware optimization techniques, and explainable artificial intelligence through Grad-CAM visualization. By combining these components, the framework was designed to achieve both strong classification performance and clinically interpretable predictions.

The experimental results demonstrate that the proposed framework is capable of effectively distinguishing stenotic and non-stenotic coronary artery conditions. The integration of weighted sampling and focal loss optimization successfully addressed the class imbalance problem commonly encountered in medical imaging datasets, improving sensitivity toward clinically significant stenosis cases. Furthermore, the use of transfer learning enabled efficient feature extraction and accelerated model convergence despite limited annotated medical data.

The Grad-CAM explainability analysis showed that the model consistently focused on anatomically relevant coronary vessel regions during prediction, indicating that the framework learned meaningful pathological representations rather than relying on irrelevant image artifacts. This interpretability aspect is essential for improving clinician trust and supporting the adoption of AI-assisted diagnostic systems in healthcare environments.

Overall, the proposed EfficientNet-B4–based framework demonstrates strong potential as a computer-aided diagnostic tool for coronary stenosis detection. The combination of high predictive capability, robust optimization strategies, and visual explainability contributes to a reliable and clinically relevant solution for assisting cardiologists in coronary angiography interpretation.

Nevertheless, several limitations remain, including limited dataset diversity and the current focus on binary classification. Future work should explore larger multi-center datasets, stenosis severity assessment, real-time localization systems, and hybrid deep learning architectures to further improve robustness and clinical applicability. Integration with real-world clinical decision support systems also represents an important direction for future research.

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